The secondary image of each star appears between the two Einstein rings. Each secondary image circulates along a closed curve (for example, the two yellow curves), and it circulates in the opposite direction to the red streaming motions outside the outer ring.

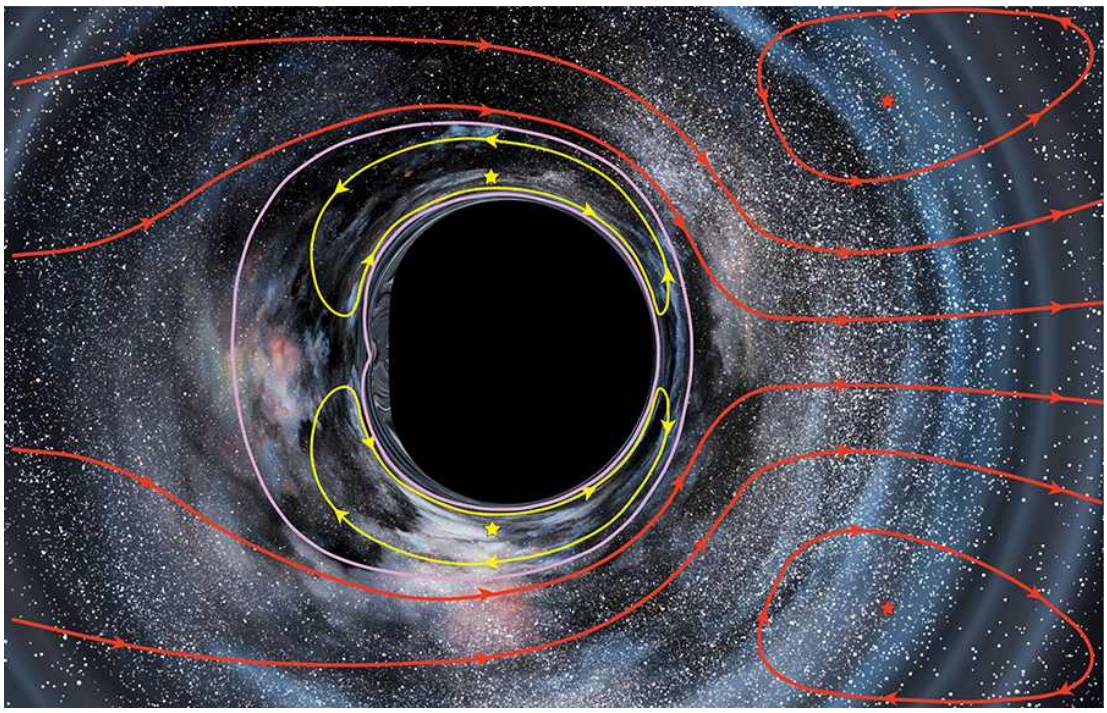


Fig. 8.5. The star streaming patterns as seen by a camera near a rapidly spinning black hole such as Gargantua. In this simulation by the Double Negative visual-effects team, the hole spins at 99.9 percent of the fastest possible, and the camera is in a circular, equatorial orbit with circumference six times larger than the horizon's circumference. For a film clip of this simulation, see this book's page at Interstellar.withgoogle.com.

There are two very special stars in Gargantua's sky with gravitational lensing turned off. One lies directly above Gargantua's north pole; the other directly below its south pole. These are analogs of the star Polaris, which resides directly above the Earth's north pole. I placed five-pointed stars at the primary (red) and secondary (yellow) images of Gargantua's pole stars. All the stars on the Earth's sky appear to circulate around Polaris as we humans are carried around by the